Temporal vs. Per-Frame Recognition: Do Models Use Gradual Facial Expression Progression to Classify Emotions?

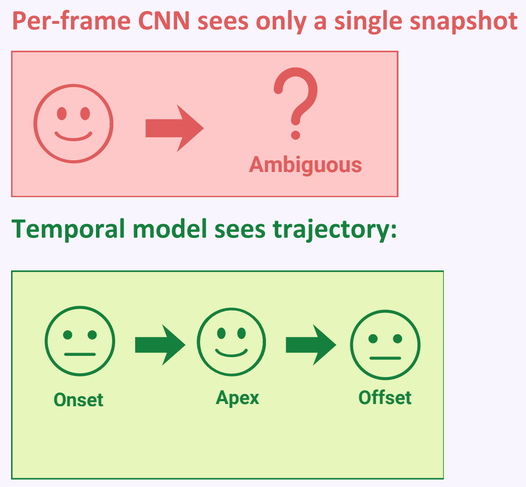
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Motivation

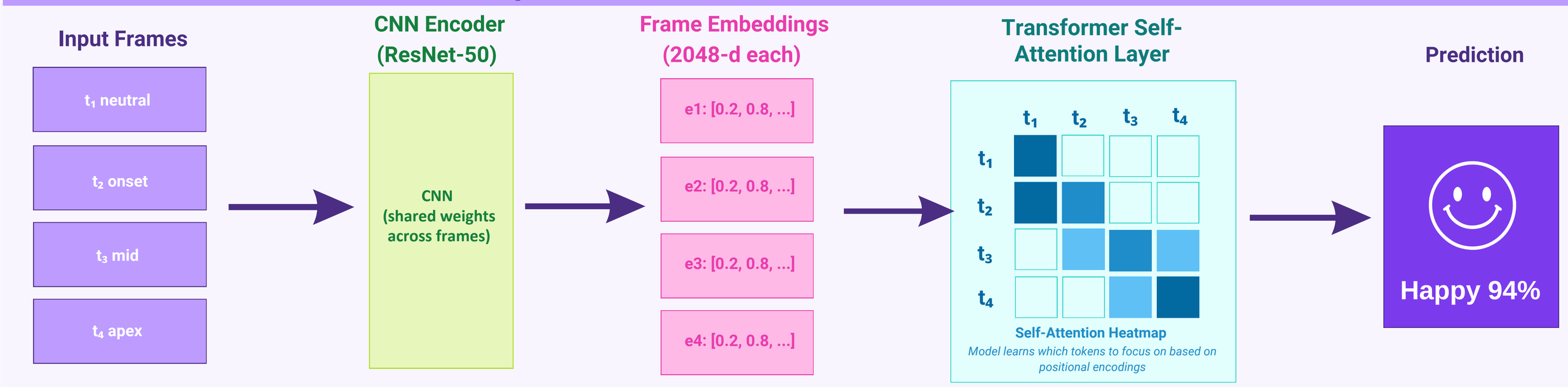
Facial expressions are not static events, they unfold over time through three distinct phases: onset (neutral → expression begins), apex (peak intensity), and offset (expression fades).

A per-frame model sees only a single snapshot. At onset or offset, that snapshot is ambiguous. For example, a slight mouth curl could be the beginning of happiness or disgust. Without knowing what came before, the model guesses. To make it more fair when comparing results, we will be averaging out the result for the per-frame CNN over the same number of frames (i.e. 15) that our CNN+Transformer model and CNN+LSTM model gets to see.

A temporal model sees the trajectory. Knowing the face was neutral two seconds ago and is now slightly asymmetric lets the model make an informed prediction about what's developing.



How CNN + Transformer Adds Temporal Context



Dataset

RAVDESS (Livingstone & Russo, 2018)

- **24 professional actors** (12F / 12M)
- **8 emotions:** neutral, calm, happy, sad, angry, fearful, disgust, surprised
- **Video-only modality** (.mp4, 720p)
- **~1,440 clips**, 3–5s each
- **Single emotion per clip** (controlled setting)

Preprocessing

- **Extract frames at 5 fps** (OpenCV)
- **Detect & crop face region** (MTCNN)
- **Resize 224×224, normalize** (ImageNet μ/σ)

Experimental Setup

Why Both LSTM and Transformers?

LSTM captures temporal context sequentially and processes frames one step at a time, while the **Transformer**'s self-attention allows the apex frame to directly compare itself with earlier neutral onset frames in a single operation. We hypothesize that this direct frame-to-frame comparison is more informative for short RAVDESS clips.

Training

- **Adam optimizer** with learning rate 1e-4 and weight decay 5e-4
- **ReduceLROnPlateau scheduler** (patience = 5, factor = 0.5)
- Trained for **20 epochs** with **batch size 16** for sequence models and **32** for the baseline
- 70/15/15 **train-validation-test split** grouped by actor ID to prevent identity leakage between splits

Model Architectures

Baseline: Per-Frame CNN

ResNet-50 is fine-tuned to classify each of the same 15 frames independently, without using temporal information. Predictions are averaged across frames so that the amount of visual information remains identical to the temporal models, isolating the effect of temporal reasoning.

CNN + LSTM

ResNet-50 embeddings → 2-layer BiLSTM (hidden=512). Sequential context via hidden state.

CNN + Transformer

ResNet-50 embeddings → Transformer Encoder (2L, 8H). Direct attention across all frames.

CNN + Transformer (No PE)

ResNet-50 frame embeddings are passed to a 2-layer, 8-head Transformer Encoder without positional encoding. The model receives multiple frames simultaneously but is not explicitly given frame order information.

Results

Model	Validation Accuracy	Test Accuracy
ResNet-50	62.1%	47%
CNN+Transformer	69.2%	53.7%
CNN+Transformer (No PE)	68.3%	57%
CNN+BiLSTM	71.2%	57.3%

Table 1. Validation and test accuracy comparison of each model

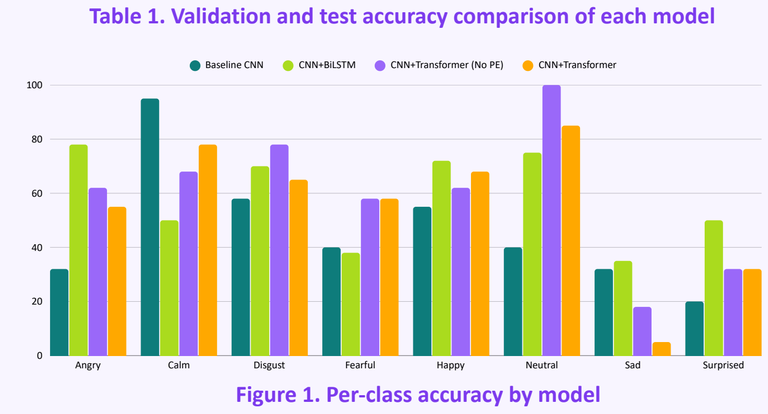


Figure 1. Per-class accuracy by model

- **Temporal models beats per frame baseline.** The ResNet-50 only model has a 10% accuracy gap from the temporal models
- **CNN+Transformer model with positional encoding (PE) underperforms the version without PE.** Highlights a limitation of PE in smaller datasets due to overfitting to different actors' styles of acting/changes in emotion
- **Val accuracy >> Test accuracy** due to checkpoint choice & ReduceLROnPlateau
- **Baseline model defaults to "calm" expression, something that's resolved in temporal models.** Temporal models resolve ambiguity with context from multiple frames ("angry" goes from 33% to 78%).
- **Disgust and neutral** are the transformer's strength.

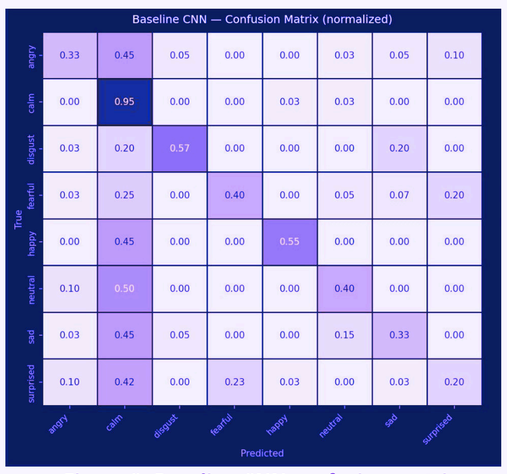


Figure 2. Baseline CNN confusion matrix

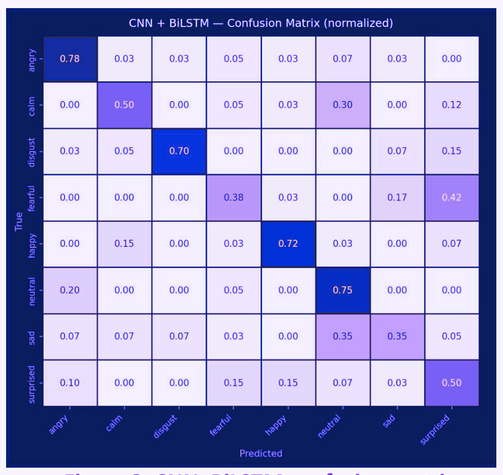


Figure 3. CNN+BiLSTM confusion matrix

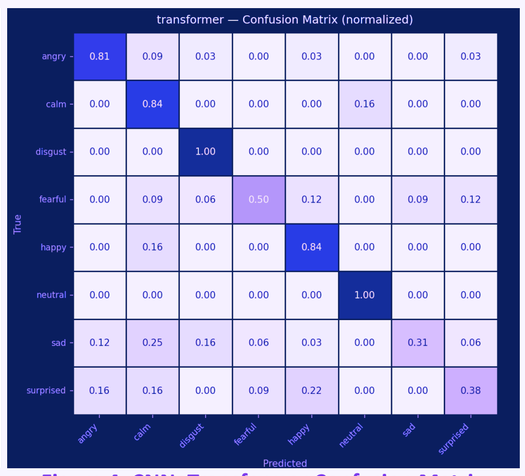


Figure 4. CNN+Transformer Confusion Matrix

Citations

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